

$$Q \Rightarrow 1024 \times 512$$

$$K^T \Rightarrow 512 \times 1024$$

$$QK^T \Rightarrow 1024 \times 1024$$

Week 11 LLM GA

512 multiply
511 add ns

1024 flops

total elements

$$1024 \times 1024 \times 1024$$

Consider a GPT model using a full attention mechanism in the attention layer. Assume the model has only one layer and a single head. Suppose the context length of the model is $T = 1024$.

Assume $d_{model} = 512$ and ($d_q = d_k = d_v = d_{model}$) and one FLOP for a single mathematical operation such as addition and multiplication.

1) Calculate the exact number of FLOPs (Floating Point Operations) required to compute the pre-attention scores. Enter your answer in MFLOPs. For example, for 373.43 MFLOPs, enter 373.43

1072.69

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Range) 1072,1073

$$\frac{1024 \times 1024 \times 1024}{10^6} = 1072.74$$

$$1073741824 \times$$

1 point

2) What is the amount of memory required to store these pre-attention scores if we assign 16 bits per number? Enter the answer in KB

524

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 2096,2098

$$\frac{2147483648}{1024} = 2097152 \text{ bytes}$$

$$= 2098 \times 1024$$

1 point

3) Suppose we want to reduce the complexity of the attention module in the encoder part of the transformer from $O(T^2)$ to at max $O(\frac{T^2}{2})$ using a local attention mechanism. Then which of the following values of c (window size) in the strided local attention helps achieve this objective? 1 point

☒ $c = 512$

☐ $c = 768$

☒ $c = 256$

☒ $c = 128$

The diagram below shows a few possible permutation blocks (for four attention heads) to be used in the masking matrix M in the block-wise attention mechanism.

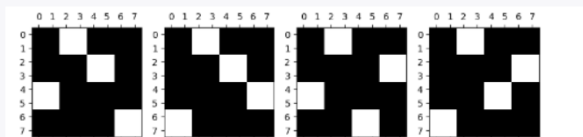


Figure 1: From left to right: a,b,c,d

4) Of these 4 blocks, choose the block(s) that is (are) proper permutation of 4 blocks.

1 point

☒ a

☒ b

☒ c

☒ d

☐ None of these

all, because there can 4! possibilities

Yes, the answer is correct.

Score: 1

5) Let q_1, q_2, q_3, q_4 denote the 4 query blocks and k_1, k_2, k_3, k_4 denote the 4 key blocks. Select which of the following denotes the computation of pre-attention score for the q_4 in the 4-th attention head **1 point**

- ☐ $q_4^T k_4$
- ☒ $q_4^T k_1$
- ☐ $q_4^T k_2$
- ☐ $q_4^T k_3$

$$\text{score}(i, j) = q_i^T k_j$$

$$q_4^T k_1$$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$q_4^T k_1$

6) The statement that both a low-rank approximation of attention and local attention has the same computational complexity when $k = 1$ and $c = 1$ and therefore both approaches always end up computing the same representation for the given tokens in a sequence **1 point**

- ☐ True
- ☒ False

fixed window
it compresses the info.

Yes, the answer is correct.

Score: 1

7) Suppose we decided to use KV caching mechanism for faster inference. The details of the text generation model are given below

1. Number of layers 12
2. Number of heads per layer 8
3. $d_{model} = 512$
4. $d_q = d_k = d_v = 64$
5. Precision 32 bits

What will be the minimum size of cache memory (in MB) such that it should cache at least 512 tokens. Enter your answer in MB, for example, enter 1.23 for 1.23 MB.

12.5

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 24,26

$$2 \times 12 \times (8 \times 64) \times 4$$

$$\Rightarrow 49152 \times 512 \Rightarrow$$

$$\frac{25,165,824}{1024 \times 1024} = 24$$

1 point

8) The statement that KV caching is not helpful during inference in encoder models like BERT is

- ☒ True
- ☐ False

Yes, the answer is correct.

Score: 1

yes, since that time all the tokens will be newly generated. This can only help to reduce the time for training time.